

SEQUENCES

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Lecture Outline

- 1 Introduction
- 2 Types of Sequences
- 3 Limit of Sequences
- 4 Theorems of Sequences
- 5 Bounded Sequences
- 6 Monotonic Sequences



Introduction

Definition

A sequence is a list of numbers in a given order:

$$U_1, U_2, \dots, U_n, \dots \quad (1)$$

- ① U_i 's are numbers and are called the **terms** of the sequence.
- ② U_i is the term of **index** i or the i 'th term.
- ③ If the sequence has a last term, then it is called a **finite sequence**.
- ④ If the number of terms is infinite, then the sequence is called an **infinite sequence**.
- ⑤ The n_{th} term of the sequence is often denoted by U_n , while the sequence itself will be denoted by $\{U_n\}$.



Introduction

Definition (Infinite Sequence)

An **infinite sequence** of numbers is a function $U: \mathbb{N} \rightarrow \mathbb{R}$, from the positive integers to the real numbers. We denote $U(n)$ by U_n .

The members of the sequence usually called **terms** gets listed as **first**, second, **third**, and so on. So we write $U_1, U_2, U_3, \dots$.

Example

- ① $U_n = n - 1$ is the sequence $0, 1, 2, 3, 4, \dots$.
- ② $U_n = (-1)^{n+1} \frac{1}{n}$ is the sequence $1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots$
- ③ $U_n = \frac{n-1}{n}$ is the sequence $0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$
- ④ $U_m = 5$ is the sequence $5, 5, 5, 5, \dots$.

Types of Sequences

Definition (Geometric Sequences)

A sequence is said to be geometric if

$$\forall n \in \mathbb{N}, \frac{U_n}{U_{n-1}} = r \quad (2)$$

where $r \in R$. The number r is called the **common ratio**.

Example

The sequence

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots \quad (3)$$

is a geometric sequence with $U_1 = 1$ and $r = \frac{1}{2}$



Types of Sequences

Arithmetic sequence

A sequence is said to be arithmetic if

$$\forall n \in \mathbb{N}, \quad U_n - U_{n-1} = d \quad (4)$$

where $d \in R$. The number d is called the **common difference**.

Example

The sequence 2, 4, 6, 8, 10, $\dots$ is an arithmetic sequence.

Note

There are sequences which are neither arithmetic nor geometric. This will be discussed later.



Limit of Sequences

Loosely speaking we say that ℓ is a **limit** of sequence if $U_n - \ell$ is “small” for sufficiently large values of n .

That is the values of n gets close to ℓ such that the distance from U_n to ℓ is very small.

Example

- ① The constant sequence

$$5, 5, 5, \dots$$

has **limit 5** as for large values of n .

- ② Similarly the sequence

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

has **limit 0**.

- ③ On the other hand sequences such as $1, 2, 3, 4, \dots$ and $1, -1, 1, -1, \dots$ have **no limit**.

Limit of Sequences

Definition

- 1 Let $\ell \in \mathbb{R}$.
- 2 Then ℓ is said to be the limit of an infinite sequence U_n if for any $\varepsilon > 0$ there is $m(\varepsilon) \in \mathbb{N}$ such that

$$|U_n - \ell| < \varepsilon \quad (5)$$

for all $n \geq m$.

- 3 We write

$$\lim_{n \rightarrow \infty} U_n = \ell$$

- 4 or

$$U_n \rightarrow \ell \text{ as } n \rightarrow \infty.$$



Definition (Convergent and Divergent Sequences)

If an infinite sequence of real numbers has a limit ℓ , we say that the sequence is **convergent**. If the sequence does not have a limit then we say that it is **divergent**.

In other words:

Definition

The sequence U_n is said to be **convergent** if $\lim_{n \rightarrow +\infty} U_n$ is finite. Otherwise, the sequence is **divergent**.

Example

The sequence $U_n = \frac{n}{n+1}$ is convergent since $\lim_{n \rightarrow +\infty} U_n = 1$.

Whereas the sequence $V_n = n \ln n$ diverges.



Theorem

If an infinite sequence of real numbers has a limit, then the limit is unique.
(In other words, a convergent sequence has only one limit).

Proof

- 1 We prove by contradiction. Assume that a convergent sequence U_n has two distinct limits ℓ_1 and ℓ_2 so that $|\ell_1 - \ell_2| > 0$. Let $\varepsilon = \frac{1}{2}|\ell_1 - \ell_2|$.
- 2 By the assumption, there is some $m_1 \in \mathbb{N}$ such $|U_n - \ell_1| < \varepsilon$ whenever $n \geq m_1$ and similarly there is some $m_2 \in \mathbb{N}$ such $|U_n - \ell_2| < \varepsilon$ whenever $n \geq m_2$.
- 3 Let $m = \max\{m_1, m_2\}$. Then $m \geq m_1$ and $m \geq m_2$ and so for $n \geq m$ both $|U_n - \ell_1|$ and $|U_n - \ell_2|$ are less than ε .



Thus

$$\begin{aligned} |\ell_1 - \ell_2| &= |(U_n - \ell_1) - (U_n - \ell_2)| \\ &\leq |(U_n - \ell_1)| + |(U_n - \ell_2)| \\ &< \varepsilon + \varepsilon \\ &< 2\varepsilon \end{aligned}$$

But $\varepsilon = \frac{1}{2}|\ell_1 - \ell_2|$ and so

$$\begin{aligned} |\ell_1 - \ell_2| &< 2 \cdot \frac{1}{2} |\ell_1 - \ell_2| \\ &< |\ell_1 - \ell_2| \end{aligned}$$

This is a contradiction. Hence $\ell_1 = \ell_2$.



Theorem

Let U_n and V_n be two sequences such that $\lim_{n \rightarrow \infty} U_n = \ell_1$ and $\lim_{n \rightarrow \infty} V_n = \ell_2$. Then the following results hold.

- a. $\lim_{n \rightarrow \infty} (U_n \pm V_n) = \ell_1 \pm \ell_2$
- b. $\lim_{n \rightarrow \infty} U_n \cdot V_n = \ell_1 \cdot \ell_2$
- c. $\lim_{n \rightarrow \infty} \left(\frac{U_n}{V_n} \right) = \frac{\ell_1}{\ell_2}$ provided $\ell_2 \neq 0$.
 - i. If $\ell_2 = 0$ and $\ell_1 \neq 0$ then $\lim_{n \rightarrow \infty} \left(\frac{U_n}{V_n} \right)$ does not exist.
 - ii. If $\ell_2 = 0$ and $\ell_1 = 0$ then $\lim_{n \rightarrow \infty} \left(\frac{U_n}{V_n} \right)$ may or may not exist.
- d. $\lim_{n \rightarrow \infty} U_n^p = \left(\lim_{n \rightarrow \infty} U_n \right)^p = \ell_1^p$
- e. $\lim_{n \rightarrow \infty} p^{U_n} = p^{\lim_{n \rightarrow \infty} U_n} = p^{\ell_1}$.



Evaluating Limits

Example

Find $\lim_{n \rightarrow \infty} \frac{2n}{4n^{\frac{1}{2}} + n}$

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{2n}{4n^{\frac{1}{2}} + n} &= \lim_{n \rightarrow \infty} \frac{\frac{2n}{n}}{\frac{4n^{\frac{1}{2}}}{n} + \frac{n}{n}} \\ &= \lim_{n \rightarrow \infty} \frac{2}{\frac{4}{n^{\frac{1}{2}}} + 1} \\ &= 2\end{aligned}$$



Theorem

Let n_0 be any positive integer and f be a function defined for all $x \geq n_0$.

If

$$u_n = f(n) \text{ for } n \geq n_0$$

then

$$\lim_{n \rightarrow \infty} u_n = \lim_{x \rightarrow \infty} f(x)$$

Theorem (Absolute Value Theorem)

If the

$$\lim_{n \rightarrow \infty} |u_n| = 0,$$

then

$$\lim_{n \rightarrow \infty} u_n = 0$$



Some special limits

$$① \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

$$② \lim_{n \rightarrow \infty} x^n = \begin{cases} 0 & : |x| < 1 \\ +\infty & : x > 1 \\ \text{does not exist} & : x \leq -1. \end{cases}$$

$$③ \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \text{ for all real number } x.$$

$$④ \left(1 + \frac{x}{n}\right)^n = e^x \text{ for all real number } x. \text{ This implies } \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$⑤ \lim_{n \rightarrow \infty} \frac{n!}{\sqrt{n}^{(n/e)^n}} = \sqrt{2\pi}. \text{ This is the } \text{Stirling's Formula}$$

$$⑥ \text{ The sequence defined by the recursion formula } F_1 = F_2 = 1 \text{ and } F_{n+2} = F_{n+1} + F_n \text{ is called Fibonacci sequence and } \lim_{n \rightarrow \infty} \frac{F_n}{F_{n-1}} = \frac{1 + \sqrt{5}}{2}.$$



Bounded Sequences

Definition

- 1 A sequence U_n is said to be **bounded above** if there is a number M such that $U_n \leq M$ for all n .
- 2 A sequence U_n is said to be **bounded below** if there is a number m such that $U_n \geq m$ for all n .
- 3 If a sequence is both bounded above and below, we say that it is bounded.



Bounded Sequences

Example

- ① The sequence $U_n = \frac{n}{n+1}$ is bounded since $0 < U_n < 1$ for all $n \in \mathbb{N}$.
- ② The sequence $U_n = \frac{1}{n}$ is bounded since $0 < U_n < 1$ for all $n \in \mathbb{N}$.
- ③ The sequence $U_n = \sqrt{1 + \frac{1}{n}}$ is bounded since $1 \leq U_n \leq \sqrt{2}$ for all $n \in \mathbb{N}$.



Monotonic Sequences

Definition

Let U_n be a sequence:

① If

$$U_n \leq U_{n+1} \quad (6)$$

then the sequence is said to be **non-decreasing**.

② Similarly the sequence is **non-increasing** if

$$U_n \geq U_{n+1} \quad (7)$$

③ A **monotonic** sequence is a sequence which is either non-decreasing or non-increasing.

Theorem

Every bounded monotonic sequence has a limit.

Example

Show that the sequence $U_n = \frac{n}{2n+1}$ is non-decreasing

We have that

$$\begin{aligned} U_{n+1} - U_n &= \frac{n+1}{2(n+1)+1} - \frac{n}{2n+1} \\ &= \frac{1}{(2n+3)(2n+1)} > 0 \quad \forall n \in \mathbb{N} \end{aligned}$$

Therefore the sequence is non-decreasing.



Remarks

In case $U_n = f(n)$, then f' and f'' tests could be used to find out the monotonicity and the boundness of U_n on a given interval.

Example

Show that the sequence $U_n = e^{n^2-4n}$ is bounded below.

pause

- $f(x) = e^{x^2-4x}$ implies
- $f'(x) = (2x-4)e^{x^2-4x}$ and $f''(x) = (2 + (2x-4)^2)e^{x^2-4x}$.
- Since $f'(x) = 0 \implies x = 2$ and $f''(2) = 2e^{-4} > 0$,
- f has a minimum value $f(2) = e^{-4}$ at $x = 2$.
- Thus, U_n is bounded below.



Exercise

① Given u_1 and/or u_2 , and the **recursion formula**, find u_3, u_4 and u_5 .

(a) $u_1 = 1, u_{n+1} = u_n + \frac{1}{2^n}$ (b) $u_1 = 2, u_2 = -1, u_{n+2} = \frac{u_{n+1}}{u_n}$.

② Prove that limit of the sequence $\frac{2n}{4n^{\frac{1}{2}} + n}$ is 2.

③ Is the sequence $U_n = \frac{1}{n} + \sin\left(\frac{1}{n}\right)$ bounded?

④ Show that the sequence $U_n = \frac{1}{n} - \frac{1}{n^2}$ is non-increasing.

⑤ Which of the sequences u_n converge or diverge?

1) $u_n = \left(1 - \frac{7}{n}\right)^n$ 2) $u_n = \sqrt[n]{4^n n}$

3) $u_n = \sqrt[n]{3^{2n+1}}$ 4) $u_n = \left(\frac{1}{n}\right)^{\frac{1}{\ln n}}$

5) $u_n = \left(\frac{3n+1}{3n-1}\right)^n$ 6) $u_n = \frac{1}{n} \int_1^n \frac{1}{x} dx$.

Exercise

Evaluate the following limits

$$① \lim_{n \rightarrow \infty} \frac{4 - 2n - 3n^2}{4n^2 + 5n}.$$

$$② \lim_{n \rightarrow \infty} \frac{1}{n}$$

$$③ \lim_{n \rightarrow \infty} \sqrt{\frac{n^2(n-1)}{1 + \frac{1}{2}n^3}}$$

$$④ \lim_{n \rightarrow \infty} \frac{\sqrt{n^2 + n} - n}{n}$$

$$⑤ \lim_{n \rightarrow \infty} \sqrt{2n^2 + 1}$$



END OF LECTURE
THANK YOU

